
Math Training — M1 Cognitive Science

Session 3 — Analysis I

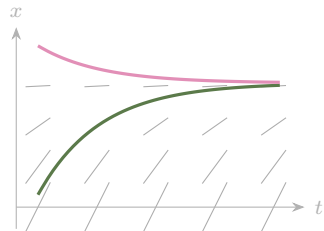
Functions, derivatives, integrals, differential equations

Advanced track

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Roadmap of the session

- Assumed: derivatives, e^{at} , the chain rule.
- Recalled in one block: antiderivatives.
- **Differential equations:** definition, geometry, existence and uniqueness.
- The linear case $\dot{x} = ax + b$, solved and proved.
- Equilibria and stability for a nonlinear f .
- Numerical solution: Euler.



Slope field of $\dot{x} = -x + 2$. **Green:** solution from $x_0 = 0.2$. **Pink:** solution from $x_0 = 2.6$.
Both approach $x^* = 2$.

■ Differentiation rules

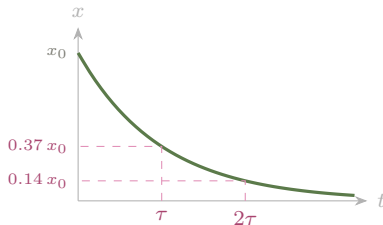
$$(t^n)' = n t^{n-1}, \quad (e^{at})' = a e^{at}, \quad (\ln t)' = \frac{1}{t}$$

$$(fg)' = f'g + fg', \quad (f \circ g)' = (f' \circ g) \cdot g'$$

■ Definition

For $a < 0$, write $a = -1/\tau$ with $\tau > 0$. The **time constant** τ is the unique time at which $e^{-t/\tau}$ has fallen to e^{-1} of its initial value.

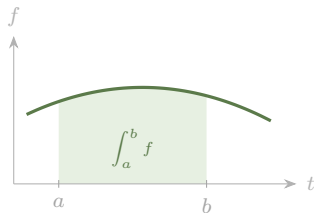
The curve is $x(t) = x_0 e^{-t/\tau}$: each further τ multiplies the remaining height by $e^{-1} \approx 0.37$ — the levels $0.37x_0$ and $0.14x_0$ are reached at $t = \tau$ and $t = 2\tau$.



■ Fundamental theorem of calculus

Let f be continuous on an interval I and $a \in I$. Then $F(t) = \int_a^t f(s) ds$ is differentiable on I with $F' = f$, and for every antiderivative G of f on I ,

$$\int_a^b f(s) ds = G(b) - G(a).$$



The green curve is f ; the shaded region from a to b has area $\int_a^b f$.

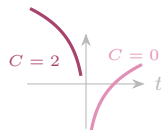
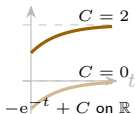
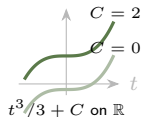
$$\int t^n dt = \frac{t^{n+1}}{n+1} \quad (n \neq -1), \quad \int \frac{dt}{t} = \ln |t|, \quad \int e^{at} dt = \frac{e^{at}}{a}$$

■ Antiderivatives differ by a constant

Two antiderivatives of f on an **interval** differ by a constant.
Three families, each solving $F' = f$:

$$\int t^2 dt = \frac{t^3}{3} + C, \quad \int e^{-t} dt = -e^{-t} + C, \quad \int \frac{dt}{t} = \ln |t| + C.$$

The first two hold with a **single** constant on all of $\mathbb{R}$. The third holds with a single constant only on an interval not containing 0 — on $(0, \infty)$ and on $(-\infty, 0)$ separately, and the two constants need not agree.

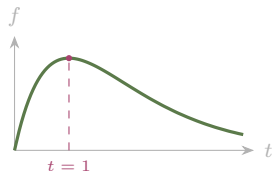


$\ln |t| + C$: two intervals, two constants

On the left branch, $\ln |t| = \ln(-t)$: it is the sign of t , not the value of C , that confines this curve to $(-\infty, 0)$ — $\ln$ simply needs a positive argument.

■ Practice — try it

1. Differentiate $f(t) = t e^{-t}$ and locate its maximum.
2. Compute $\int_0^{\infty} e^{-3t} dt$ and $\int \frac{2t}{1+t^2} dt$.
3. Compute $\int_0^{\infty} t e^{-t} dt$ by parts.



$f(t) = t e^{-t}$: one maximum, at $t = 1$.

Throughout, $R(t) = R_0 e^{-t/\tau}$ denotes the proportion of a memorised list still recalled at time t : $R' = -R/\tau$ says that the rate of forgetting is proportional to what remains.

$$\int_0^{\infty} e^{-3t} dt = \left[-\frac{1}{3}e^{-3t} \right]_0^{\infty} = \frac{1}{3}.$$

Reading: this is $\int_0^{\infty} e^{-t/\tau} dt = \tau$ with $\tau = 1/3$ — the area under the curve equals its own time constant.

$$\int \frac{2t}{1+t^2} dt \stackrel{u=1+t^2}{=} \int \frac{du}{u} = \ln|u| + C = \ln(1+t^2) + C.$$

Check: $\frac{d}{dt} \ln(1+t^2) = \frac{2t}{1+t^2}. \checkmark$

By parts, with $u = t$, $v' = e^{-t}$ (so $v = -e^{-t}$):

$$\int_0^{\infty} t e^{-t} dt = \left[-t e^{-t} \right]_0^{\infty} + \int_0^{\infty} e^{-t} dt = 0 + 1 = 1.$$

e^{-t} integrates to 1 on $[0, \infty)$, so it is a probability density, and this computation is its **mean**.

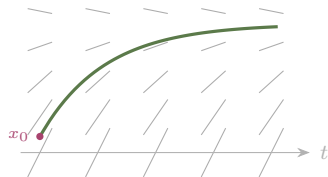
■ Definition

Let $U \subseteq \mathbb{R} \times \mathbb{R}$ be open and $f : U \rightarrow \mathbb{R}$. A **first-order scalar ODE** is

$$\dot{x} = f(t, x).$$

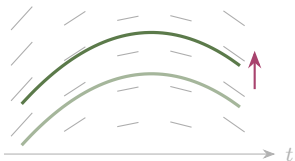
A **solution** is a pair (I, x) : I a non-degenerate interval, $x : I \rightarrow \mathbb{R}$ differentiable, with $(t, x(t)) \in U$ and $\dot{x}(t) = f(t, x(t))$ for every $t \in I$.

The equation is **autonomous** if f does not depend on t . The **Cauchy problem** adds $x(t_0) = x_0$.



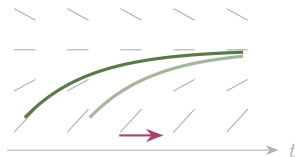
Slope field of $\dot{x} = -x + 2$. Green: the solution with $x(0) = x_0 = 0.25$ — its tangent matches the segment drawn at every point it visits.

$\dot{x} = f(t)$ and $\dot{x} = f(x)$ are geometrically different



slope constant on **vertical** lines

On the left: $\dot{x} = f(t)$ — an integration problem, $x(t) = x_0 + \int_{t_0}^t f(s) ds$. Shifting a solution **up** gives another solution.



slope constant on **horizontal** lines

On the right: $\dot{x} = f(x)$, autonomous — if x solves it, so does $t \mapsto x(t + c)$ for every c . Shifting a solution **sideways** gives another solution.

A solution of $\dot{x} = -(x - 3)$ is **not** $\varphi(x) = -x^2/2 + 3x$.

Type argument. An antiderivative in x is a function of x ; a solution is a function of t , on an interval (previous frame). φ carries neither a t nor an interval of t -values — the two objects do not live in the same set, so φ cannot be a solution.

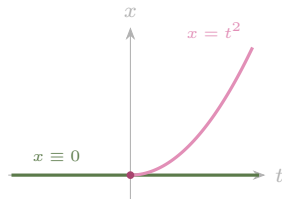
Substitution test. With $\varphi(x) = -x^2/2 + 3x$, the chain rule gives $\frac{d}{dt}\varphi(x(t)) = (3 - x)\dot{x}$, which equals $3 - x$ only if $\dot{x} = 1$ — not the equation asked.

■ Cauchy–Lipschitz, local form

Let U be open, $f : U \rightarrow \mathbb{R}$ continuous and **locally** Lipschitz in x . Then for every $(t_0, x_0) \in U$ the Cauchy problem has a **unique maximal solution**, defined on an open interval.

f of class $C^1 \Rightarrow$ locally Lipschitz in x .

Uniqueness $\Rightarrow$ **two distinct integral curves never meet.**



$\dot{x} = 2\sqrt{|x|}$, $x(0) = 0$:
uniqueness **fails**.

■ Definition

(J, y) **extends** (I, x) if $I \subseteq J$ and $y|_I = x$.

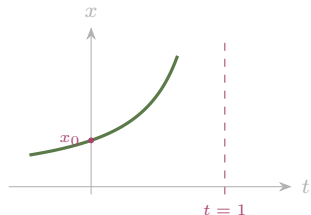
A solution is **maximal** if its only extension is itself.

■ Existence of a maximal solution

Under the hypotheses above, every solution extends to a unique maximal one, and its interval is **open**.

Why open: including a finite endpoint would let Cauchy–Lipschitz extend the solution further — contradicting maximality.

Linear equations never do this: their maximal solutions are global.



Plotted: the solution $x(t) = \frac{1}{1-t}$ of $\dot{x} = x^2$ with $x(0) = 1$, on its maximal interval $(-\infty, 1)$.

The homogeneous linear equation $\dot{x} = ax$

■ Solution of the homogeneous equation

For every $a \in \mathbb{R}$ and $x_0 \in \mathbb{R}$, the Cauchy problem $\dot{x} = ax$, $x(0) = x_0$ has the unique solution $x(t) = x_0 e^{at}$ on all of $\mathbb{R}$.

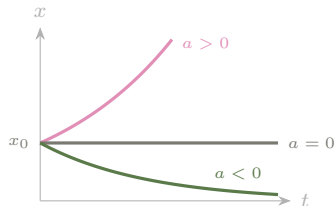
Proof. It is a solution: differentiate.

It is the only one: let y be any solution and put $z = y e^{-at}$. Then

$$\dot{z} = \dot{y} e^{-at} - a y e^{-at} = (ay - ay) e^{-at} = 0,$$

so z is constant on $\mathbb{R}$, equal to $y(0) = x_0$. $\square$

The sign of a decides the qualitative behaviour; its size only decides the speed, through $\tau = 1/|a|$.



Solutions of $\dot{x} = ax$ from the same x_0 , for $a > 0$, $a = 0$, $a < 0$.

The affine equation $\dot{x} = ax + b$

■ Solution of the affine equation

Let $a \neq 0$ and $x^* = -b/a$. The unique solution of $\dot{x} = ax + b$, $x(0) = x_0$ is

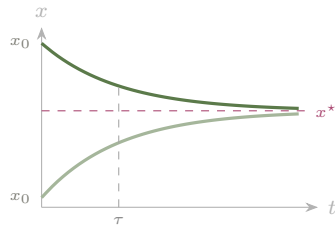
$$x(t) = x^* + (x_0 - x^*)e^{at}.$$

Proof. Put $u = x - x^*$. Then

$\dot{u} = \dot{x} = a(u + x^*) + b = au$, since $ax^* + b = 0$. Apply the previous lemma. $\square$

If $a = 0$ there is no equilibrium and $x(t) = x_0 + bt$.

Structure to remember: (particular solution) + (solution of the homogeneous equation).



Two solutions of $\dot{x} = ax + b$, from x_0 above and below x^* : one τ closes 63% of the gap.

■ Practice — try it

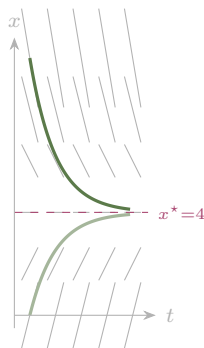
1. Classify and solve with $x_0 = 1$: (i) $\dot{x} = 3t^2$; (ii) $\dot{x} = -2x$; (iii) $\dot{x} = -2x + 5$.
2. For $\dot{x} = -x + 4$: give x^* and τ , sketch the slope field, and draw the solutions from $x_0 = 10$ and from $x_0 = 0$.
3. Does $\dot{x} = 2\sqrt{|x|}$ with $x(0) = 1$ have a unique solution near $t = 0$? And with $x(0) = 0$? Which hypothesis fails, and where?
4. Recall obeys $\dot{R} = -R/3$ with $R_0 = 0.8$ (t in days). Give $R(t)$ and $R(6)$.

Practice — linear equations, worked (I)

1. (i) $\dot{x} = 3t^2$ is an integration: $x = t^3 + 1$, on $\mathbb{R}$.
(ii) $\dot{x} = -2x$: $x = e^{-2t}$, on $\mathbb{R}$. (iii) $\dot{x} = -2x + 5$:
 $x^* = \frac{5}{2}$, so $x = \frac{5}{2} - \frac{3}{2}e^{-2t}$. Check: $\dot{x} = 3e^{-2t}$ and
 $-2x + 5 = 3e^{-2t}$. $\checkmark$

2. $x^* = 4$, $\tau = 1/|-1| = 1$, and $a = -1 < 0$:

stable. From $x_0 = 10$ the solution decreases towards 4; from $x_0 = 0$ it increases towards 4. Neither ever reaches 4: $x \equiv 4$ is itself a solution, and by uniqueness no other integral curve may cross it.



Slope field of $\dot{x} = -x + 4$: solutions from $x_0 = 10$ and $x_0 = 0$.

3. At $x_0 = 1$: **yes**, a unique solution near $t = 0$. Near $x = 1$, the map $x \mapsto 2\sqrt{x}$ is C^1 (its derivative $1/\sqrt{x}$ is continuous there), hence locally Lipschitz, and Cauchy–Lipschitz applies. At $x_0 = 0$: **no**. The Lipschitz ratio

$$\frac{|f(h) - f(0)|}{h} = \frac{2\sqrt{h}}{h} = \frac{2}{\sqrt{h}} \xrightarrow{h \rightarrow 0^+} +\infty,$$

so the hypothesis fails **at that single point**; indeed both $x \equiv 0$ and $x = t^2$ (for $t \geq 0$) solve the same Cauchy problem.

4. $R(t) = 0.8 e^{-t/3}$, so $R(6) = 0.8 e^{-2} \approx 0.108$.

■ Definition

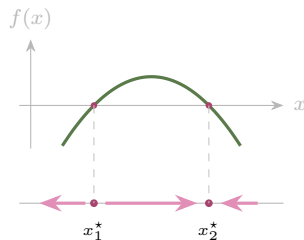
An **equilibrium** of $\dot{x} = f(x)$ is a point x^* with $f(x^*) = 0$; the constant function $x \equiv x^*$ is then a solution. It is **asymptotically stable** if every solution starting close enough converges to it.

■ Linearised stability criterion

Let f be C^1 with $f(x^*) = 0$. Then $f'(x^*) < 0$ gives a **stable** equilibrium and $f'(x^*) > 0$ an **unstable** one; if $f'(x^*) = 0$, linearisation decides nothing.

Proof idea: put $\delta = x - x^*$; then

$$\dot{\delta} = f'(x^*) \delta + o(\delta).$$



A generic f with equilibria $x_1^* < x_2^*$; below, the **sign of f** on the axis.

■ Variation of constants

Let $a \in \mathbb{R}$ and b continuous. The unique solution of $\dot{x} = ax + b(t)$, $x(0) = x_0$, is

$$x(t) = e^{at}x_0 + \int_0^t e^{a(t-s)} b(s) ds.$$

Proof. Put $y = x e^{-at}$. Then $\dot{y} = (\dot{x} - ax) e^{-at} = b(t) e^{-at}$, which is an *integration* problem: $y(t) = x_0 + \int_0^t e^{-as} b(s) ds$. Multiply back by e^{at} . □

Reading: the state at time t is the whole past input, each instant s weighted by $e^{a(t-s)}$ — exponentially discounted by how long ago it arrived. With $a < 0$ that is a leaky integrator.

■ Definition

Fix $\Delta t > 0$ and set $t_n = n\Delta t$. The **explicit Euler scheme** is

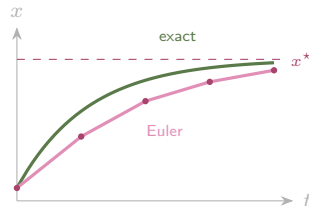
$$x_{n+1} = x_n + \Delta t f(t_n, x_n).$$

■ Error and stability of the scheme

Local error $O(\Delta t^2)$ per step, global error $O(\Delta t)$ — in practice, **halving Δt roughly halves the total error**. For $\dot{x} = ax$ the scheme gives $x_n = (1 + a\Delta t)^n x_0$, bounded iff $|1 + a\Delta t| \leq 1$, i.e. for $a < 0$ iff

$$\Delta t \leq 2/|a| = 2\tau.$$

Stability is not accuracy: staying below 2τ only stops the scheme exploding.



Exact solution of $\dot{x} = a(x - x^*)$ (green) vs. the Euler polygon (pink), from the same x_0 .

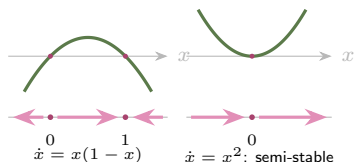
■ Practice — try it

1. $\dot{x} = x(1 - x)$: find the equilibria, classify them with f' , and draw the phase line. What happens to a solution starting at $x_0 = 0.1$? At $x_0 = -0.1$?
2. $\dot{x} = x^2$: find the equilibrium and show that f' decides nothing there. Use the phase line instead.
3. Take one Euler step of $\dot{x} = -x + 2$ from $x_0 = 0$ with $\Delta t = 0.1$. What is the largest step that does not blow up?
4. Two solutions of the same autonomous equation satisfy $x(0) = 1$ and $y(2) = 1$. How are they related?

1. $f(x) = x(1 - x)$ vanishes at $x = 0$ and $x = 1$; $f'(x) = 1 - 2x$, so $f'(0) = 1 > 0$ (**unstable**) and $f'(1) = -1 < 0$ (**stable**). From $x_0 = 0.1$ the solution rises to 1; from $x_0 = -0.1$ it decreases without bound, and in fact reaches $-\infty$ in *finite* time.

2. $f(x) = x^2$ has $x^* = 0$ and $f'(0) = 0$: the criterion is silent. But $f \geq 0$ everywhere, so the flow points **right on both sides**: attracting from the left, repelling on the right — a **semi-stable** equilibrium.

3. $x_1 = x_0 + \Delta t f(x_0) = 0 + 0.1 \times (-0 + 2) = 0.2$. Here $a = -1$, so the scheme is stable for $\Delta t \leq 2/|a| = 2$. 4. $y(t) = x(t - 2)$: the same trajectory, shifted in time — the autonomous invariance from the slope-field frame.



Graphs of f , with the sign of f carried to the phase line below each.

- An ODE is $\dot{x} = f(t, x)$; a **solution** is a function on an interval. When f is locally Lipschitz, a **Cauchy problem** has one **maximal** solution — and that interval can be finite.
- Geometrically, the equation is a **slope field** and a solution is a curve tangent to it. Distinct integral curves never meet.
- $\dot{x} = ax + b$ has the unique solution $x^* + (x_0 - x^*)e^{at}$, with $x^* = -b/a$ and $\tau = 1/|a|$.
- For nonlinear f : equilibria are the zeros of f , and the sign of $f'(x^*)$ decides stability.

Session 4: the same four statements with $x \in \mathbb{R}^n$ — $\dot{x} = \mathbf{A}x$, eigenvalues in place of a , and the Jacobian in place of $f'(x^*)$.